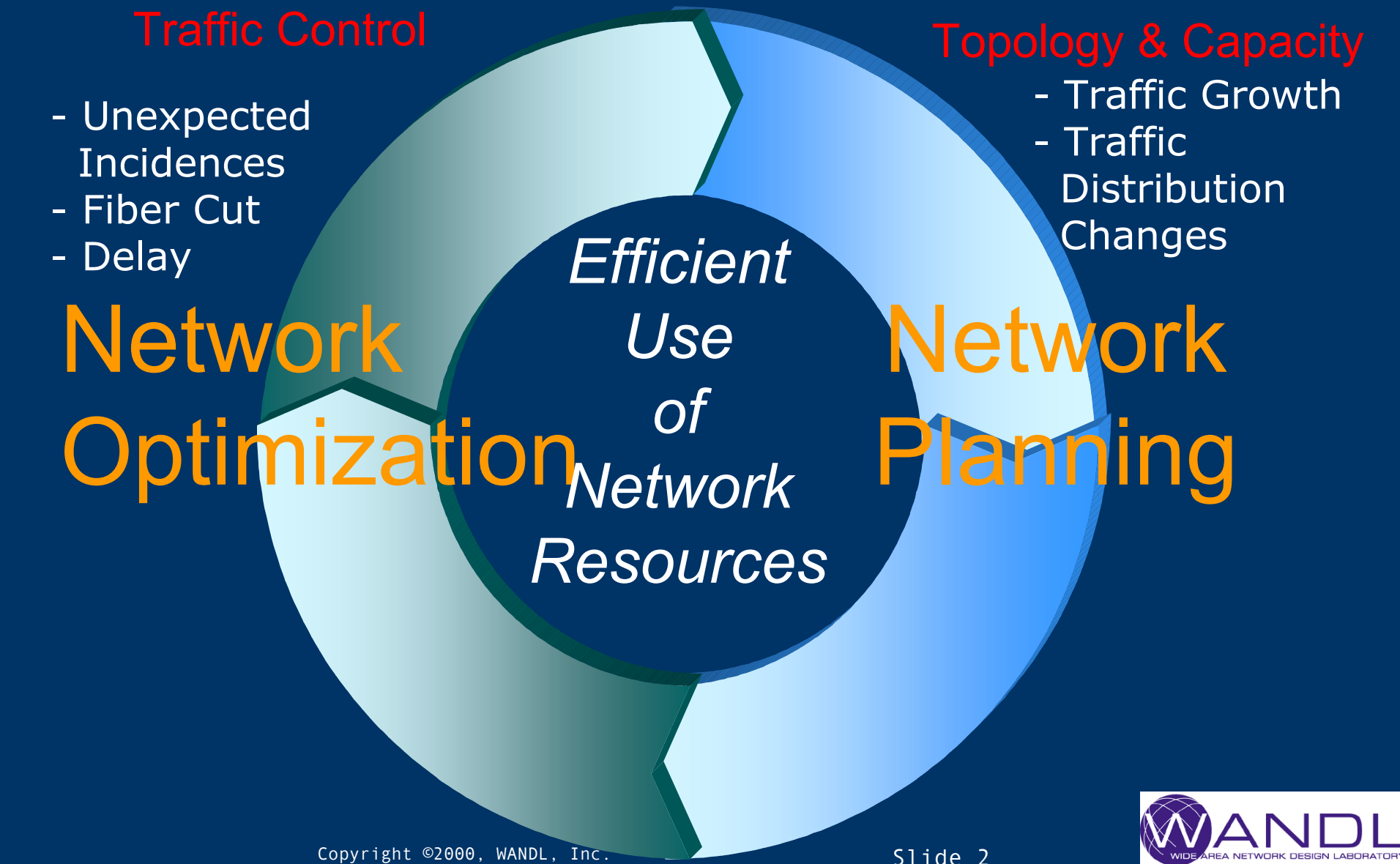


MPLS Network Design & Monitoring

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WANDL
WIDE AREA NETWORK DESIGN LABORATORY

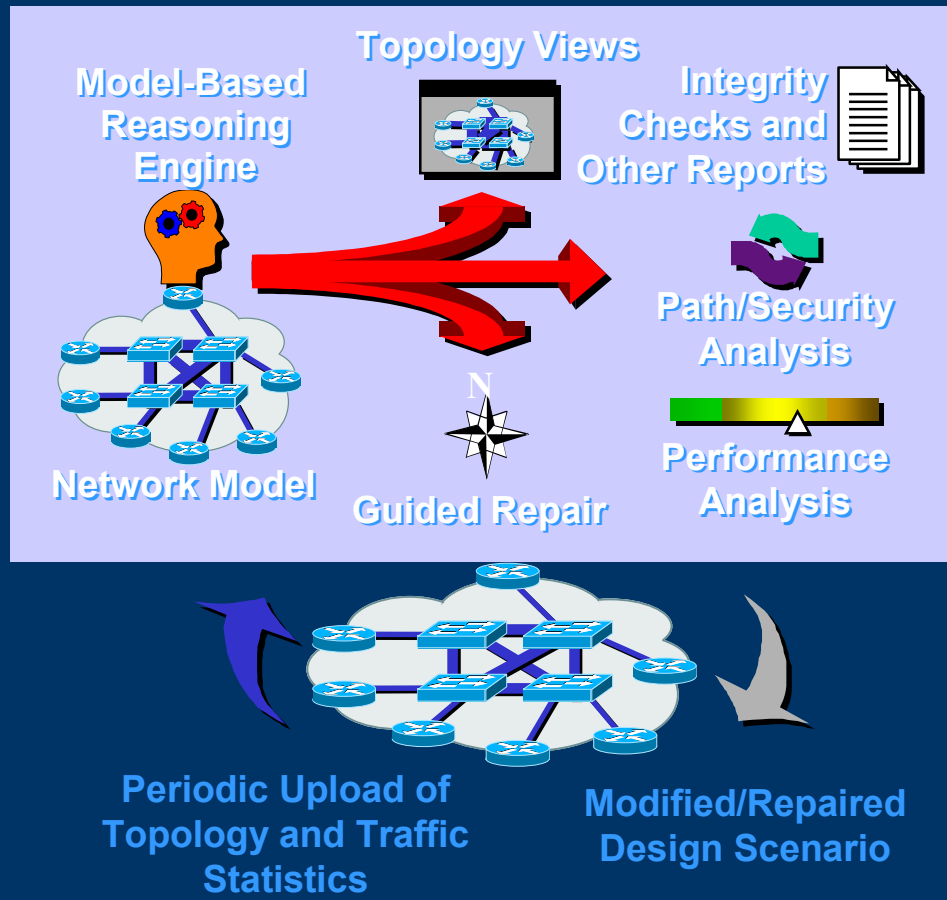
What Is MPLS Traffic Engineering?



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IP/MPLS Network Design and Traffic Engineering

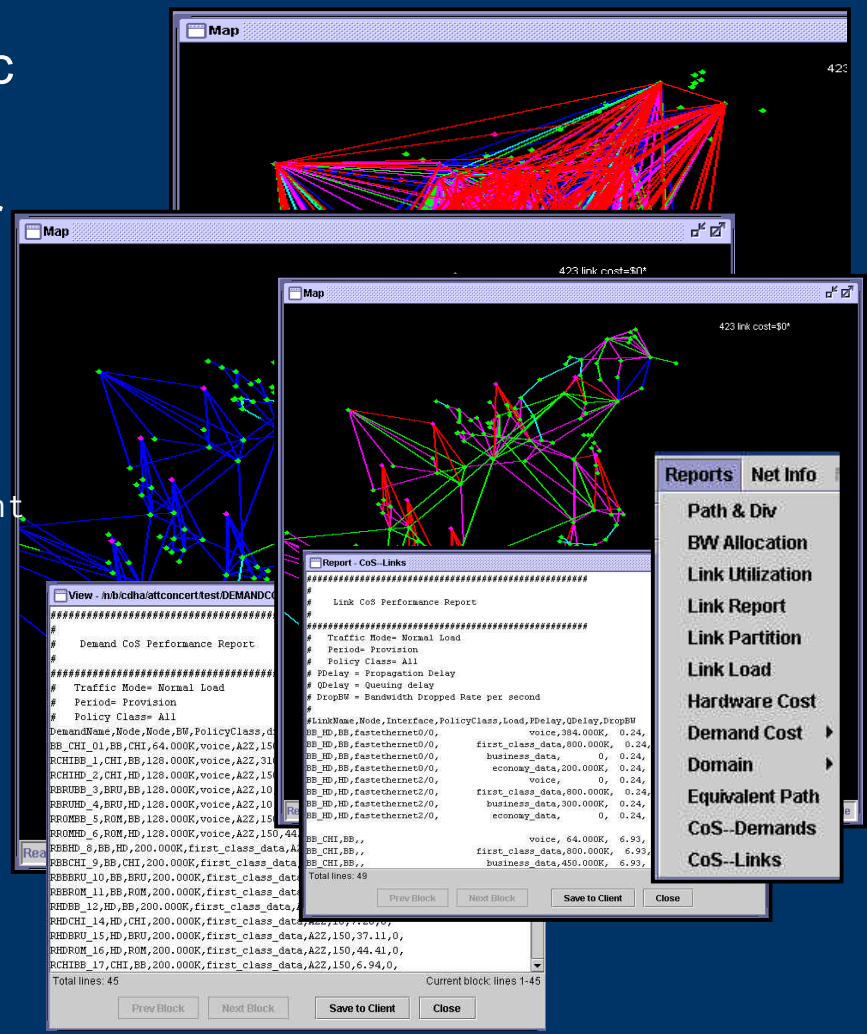
- Multi-layer routing and network planning
- Design and simulate IP/MPLS networks
- MPLS network discovery
- Multi-vendors Config files, topology data collection, parsing and integrity checking
- Planning for traffic growth
- Play “what-if” scenarios to ensure reliability of your network under stress



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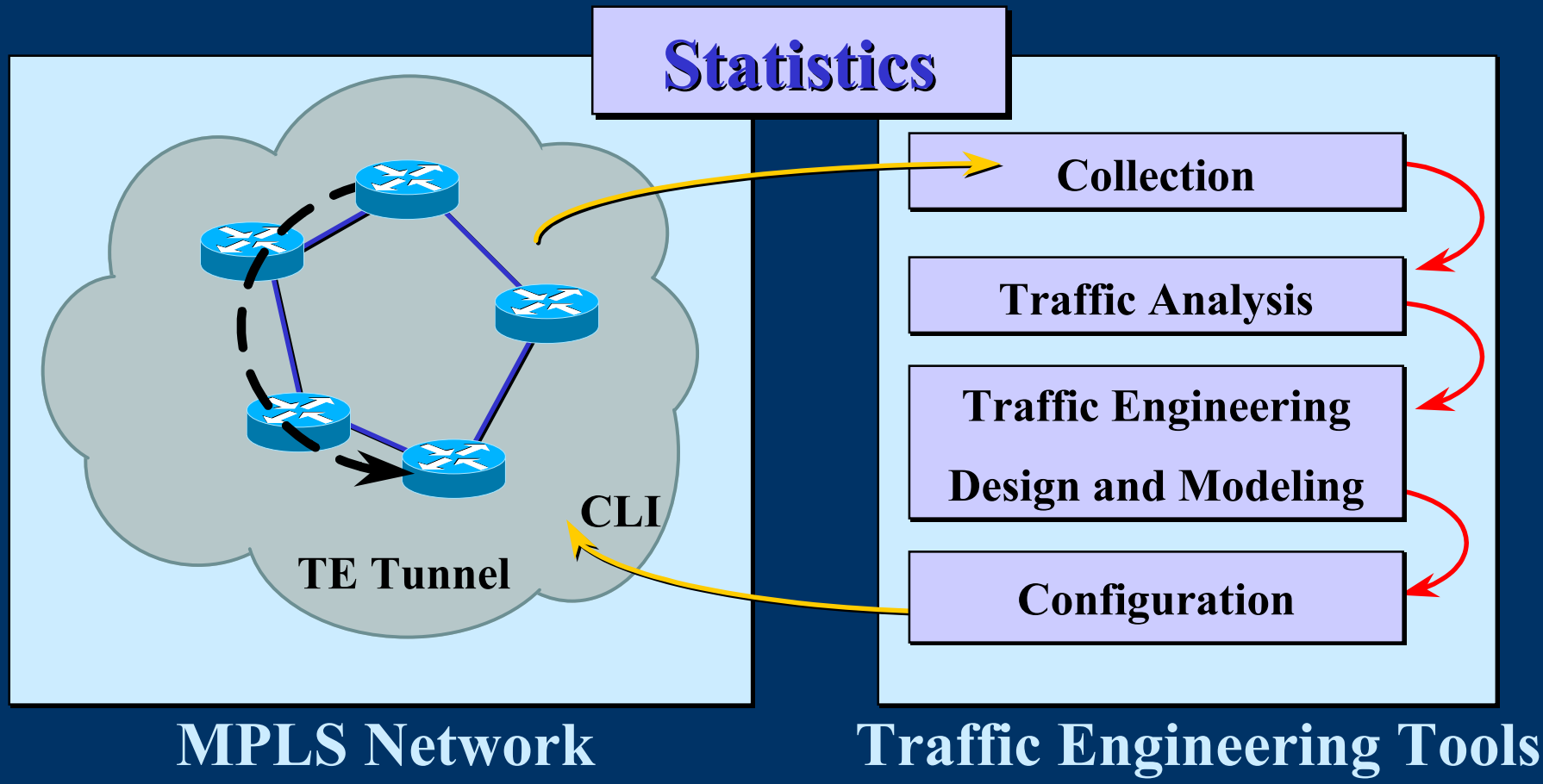
IP/MPLS Network Design and Traffic Engineering

- LSP tunnel config data and traffic collection
- Delta configuration generation for MPLS TE provisioning
- Integrated set of monitoring, modeling, and design capabilities
  - Multi-layer simulation with equipment details
  - Multi-period traffic load analysis
  - LSP tunnel path placements and provisioning
  - Offline LSP Calculation
  - What-if (fiber cut, for example) simulation
  - Primary/backup LSP path computation



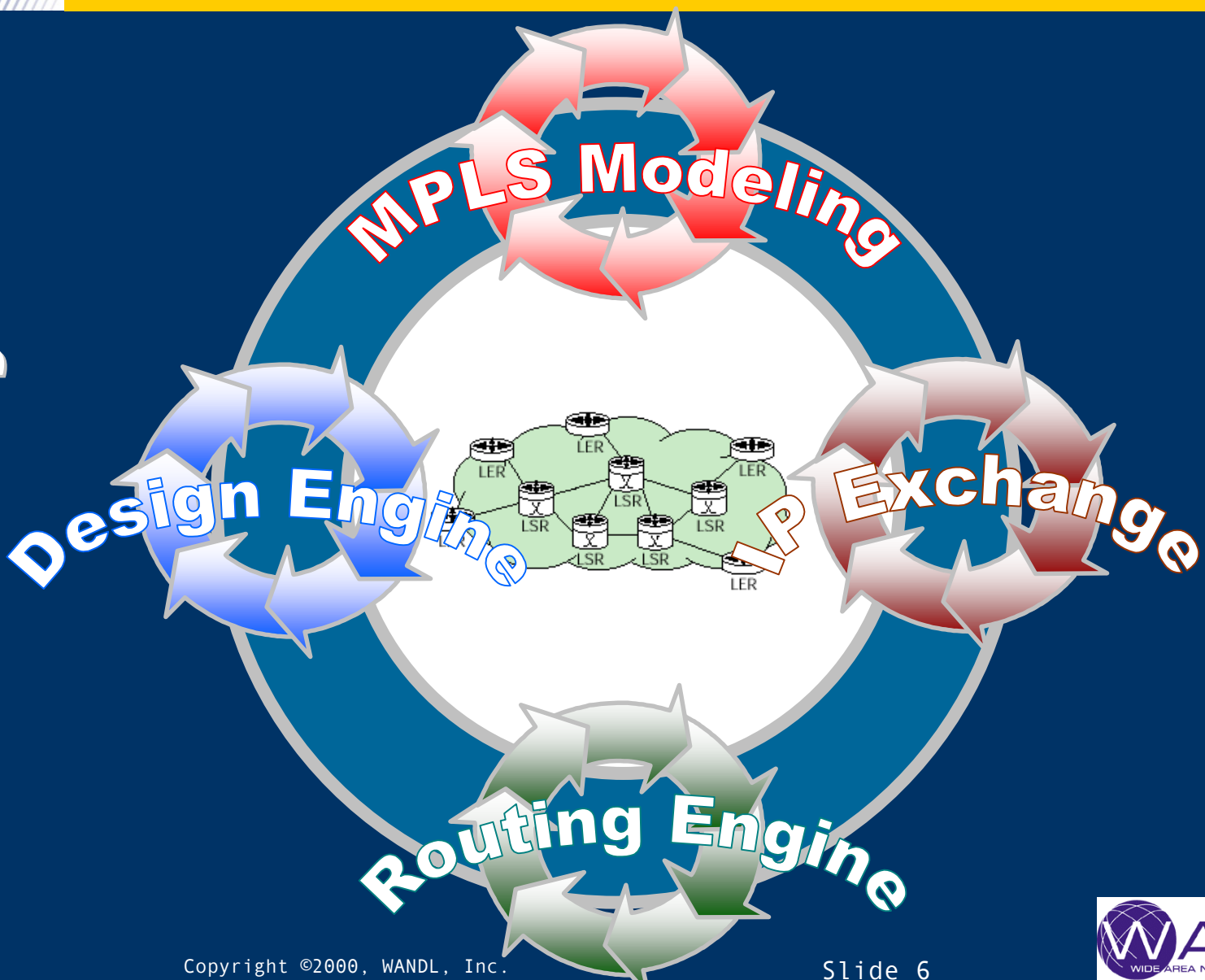
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The Traffic Engineering System



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MPLSView : *One Integrated View*



Multi-Layer Modeling

Cisco & Juniper

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WANDL MPLS Modeling



Cisco
&
Juniper

- Nodes
  - Cisco, Juniper, Foundry, Riverstone
- Links
  - T3/E3, OC3, OC12, OC48, OC192
  - 32 Bits Resource or Media Attribute
  - Routing Metrics
- LSP Tunnels
  - Source, Destination
  - Bandwidth, Affinity, Mask, Admin Group
  - Priority/Holding Priority
  - Guaranteed or Diff Serv TE, Best Effort
  - Routing Method
    - Dynamic Route
    - Explicit Route (Loose or Strict)
    - Secondary Routes
- Layer 3 Traffic Routing
  - OSPF, EIGRP, IS-IS, BGP4
  - Forwarding Equivalence Class
  - Influence of LSP Tunnels

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WANDL Data Communication



Cisco & Juniper

- Automatic Discovery (mpls, ospf, bgp)
- IOS and JUNOS Parsing
  - show config
  - show mpls traff eng tunnels
  - show ted database extensive
  - show mpls lsp detail
- Connectivity Checking
  - Pinpoint network integrity issues
- Configuration Change Management
- LSP Traffic Monitoring
- Topology Views
  - OSPF, BGP, MPLS

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MPLS Topology Parsing

MPLSView

Cisco Systems

Juniper Networks

MPLSView

File Edit Options Send Receive Window Help

```
show mpls traf topology
My_System_id: 0620.0803.2019.00, Globl Link Generation 86721

IGP Id: 0620.0803.2017.00, MPLS
      link[0 ]:Intf Address: 162
                  Nbr IGP Id: 06
          admin_weight:1, affini
          max_link_bw:2488000 ma
               allocated   reser
               ---------   -----
        bw[0]: 0           24880
        bw[2]: 0           24880
        bw[4]: 0           24880
        bw[6]: 299040      21889

      link[1 ]:Intf Address: 162
                  Nbr IGP Id: 06
          admin_weight:1, affini
          max_link_bw:2488000 ma
               allocated   reser
               ---------   -----
        bw[0]: 0           24880
        bw[2]: 0           24880
"cr1.ams2.TOPO" [Incomplete last
```

ANSI TCP/IP 21:15

MPLSView

File Edit Options Send Receive Window Help

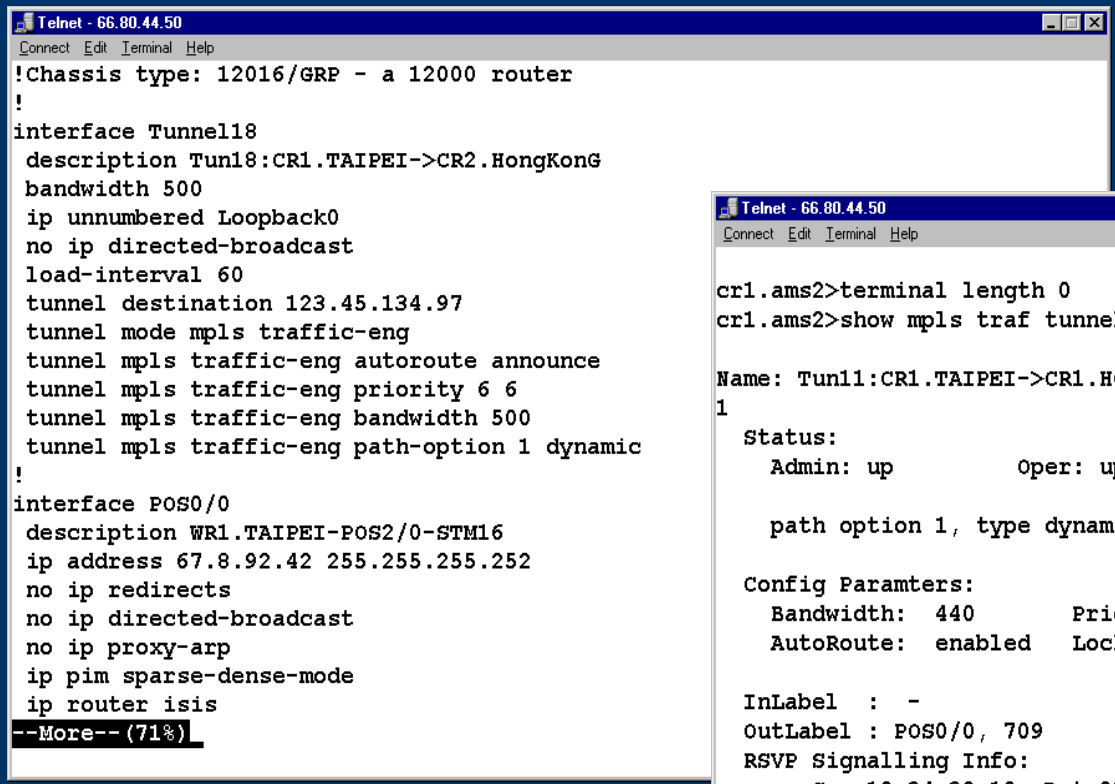
```
show ted database extensive

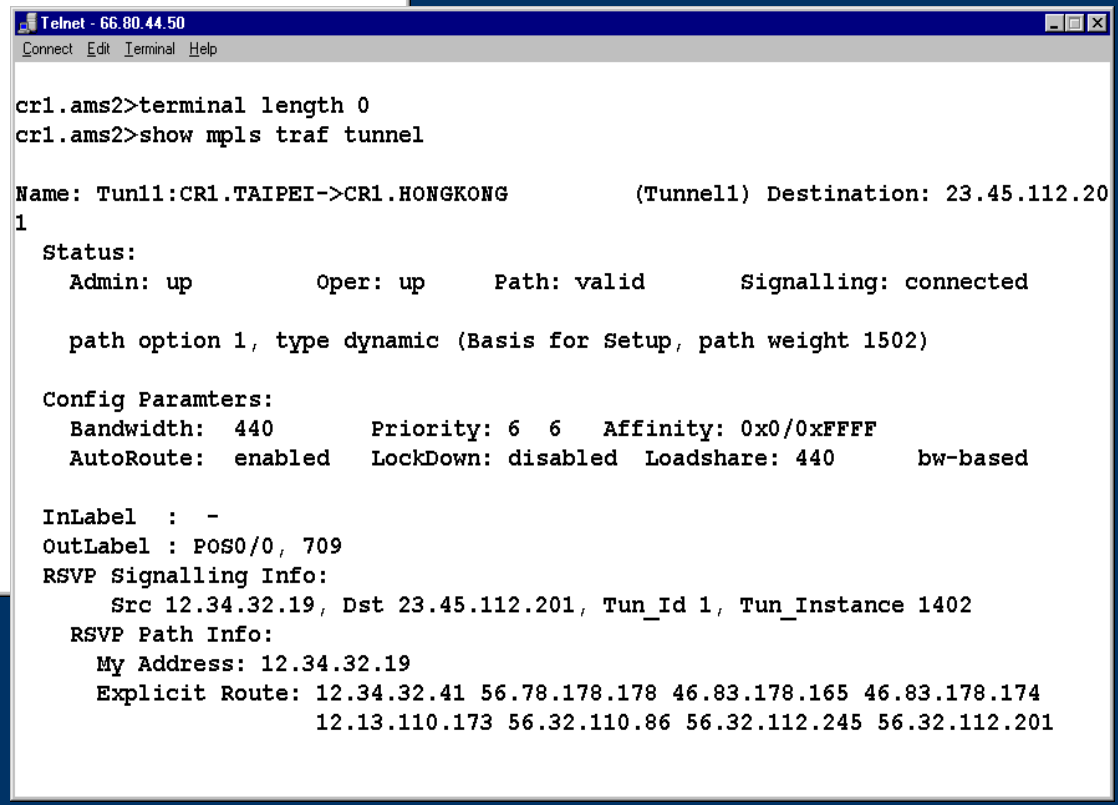
TED database: 427 ISIS nodes 256 INET nodes
NodeID: LSR_11(162.18.32.17)
  Type: Rtr, Age: 9359 secs, LinkIn: 6, LinkOut: 6
  Protocol: IS-IS(2)
    To: LSR_2.(164.28.278.146), Local: 164.28.278.177, Remote: 164.208.178.17
      Color: 0 <none>
      Metric: 600
      Static BW: 2.488Gbps
      Reservable BW: 2.488Gbps
      Available BW [priority] bps:
        [0] 2.488Gbps   [1] 2.488Gbps   [2] 2.488Gbps   [3] 2.488Gbps
        [4] 2.488Gbps   [5] 2.488Gbps   [6] 1.68684Gbps [7] 1.68684Gbps
    To: LSR_4(162.16.32.17), Local: 162.16.32.82, Remote: 162.16.32.811
      Color: 0 <none>
      Metric: 200
      Static BW: 2.488Gbps
      Reservable BW: 2.488Gbps
      Available BW [priority] bps:
        [0] 2.488Gbps   [1] 2.488Gbps   [2] 2.488Gbps   [3] 2.488Gbps
        [4] 2.488Gbps   [5] 2.488Gbps   [6] 2.02068Gbps [7] 2.02068Gbps
    To: LSR_21(162.18.36.29), Local: 162.18.36.41, Remote: 62.8.32.42
```

ANSI TCP/IP 21:27

Copyri

Cisco Router Config & Tunnel Parsing





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Juniper Router Config & Tunnel Parsing

Terminal

Window Edit Options Help

```
^G

wandl@ATL> show configuration
version 4.2R1.3;
system {
    host-name ATL;
    location npa-nxx 732868;
    root-authentication {
        encrypted-password "$1$7A2nM$idbLZHOohCiZeKv1jLn2.
    }
    login {
        class test1 {
            permissions all;
        }
        user wandl {
            full-name wandl-inc;
            uid 102;
            class test1;
            authentication {
                encrypted-password "$1$dtdnM$QfEO3K6yi.Sg
ATA
            }
        }
"10.20.0.1.cfg" [Incomplete last line] 266 lines, 5578 cha
```

Terminal

Window Edit Options Help

```
^G

wandl@SFO> show mpls lsp statistics extensive

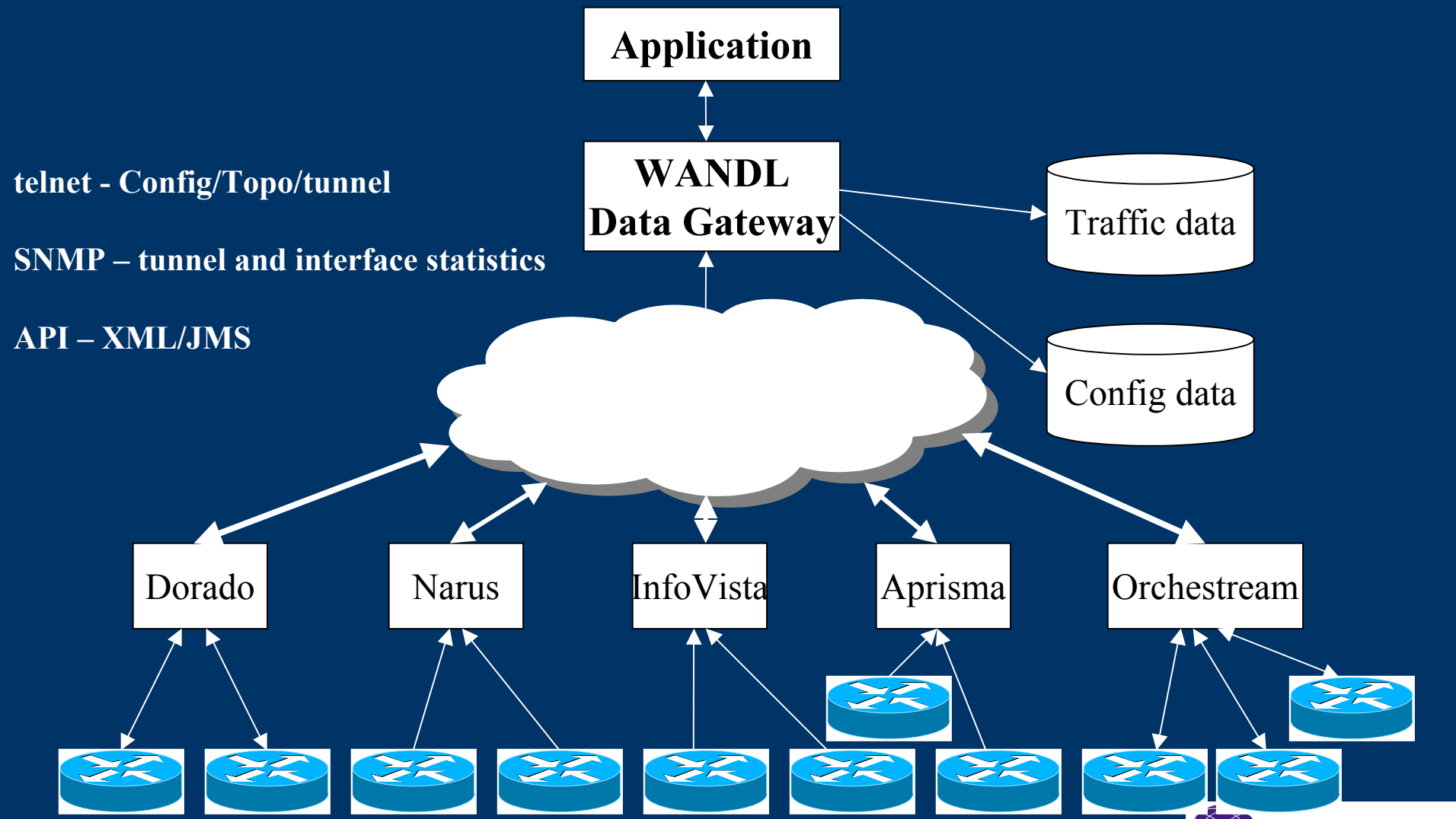
Ingress LSP: 7 label-switched paths

10.10.0.1
  From: 10.40.0.1, State: Dn, ActiveRoute: 0, LSPname: SFO2NWK
  No statistics available
  ActivePath: (none)
  LoadBalance: Random
  Metric: 1
  Primary   s2nwk              State: Dn
    3 Oct 18 17:02:16 10.0.5.1: No Route toward dest[15630 times]
    2 Sep  7 00:14:01 No Route[8 times]
    1 Sep  7 00:11:08 Originate Call
  Created: Fri Sep  7 00:11:07 2001

10.10.0.1
  From: 10.40.0.1, State: Dn, ActiveRoute: 0, LSPname: SFO2NWK2
  No statistics available
  ActivePath: (none)
  LoadBalance: Random
"10.40.0.1.lsp" [Incomplete last line] 266 lines, 10326 characters
```

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WANDL Data Communications Gateway



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LSP Routing Engine

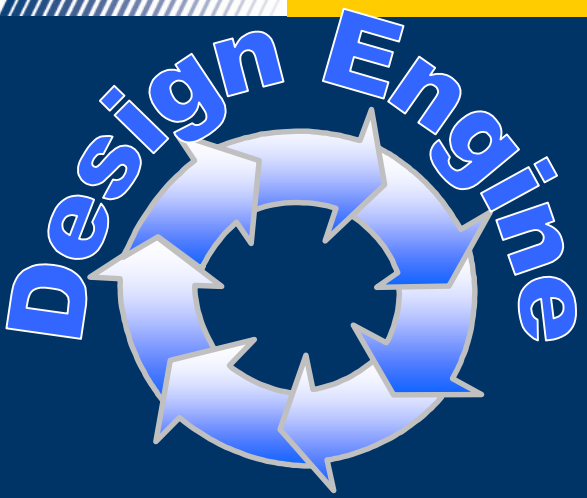


Cisco & Juniper

- LSP Routing Engine that considers:
  - All link resource constraints
  - All ingress to egress traffic trunks
- Benefits
  - similar to mechanisms in overlay nets
  - Global resource optimization
  - Predictable LSP Placement
  - Stability
  - Decision Support System
- Off-Line & Quasi On-Line Path Calculation
- Capacity Planning
- Network Grooming
- Bottlenecks Analysis

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Network Design & Optimization

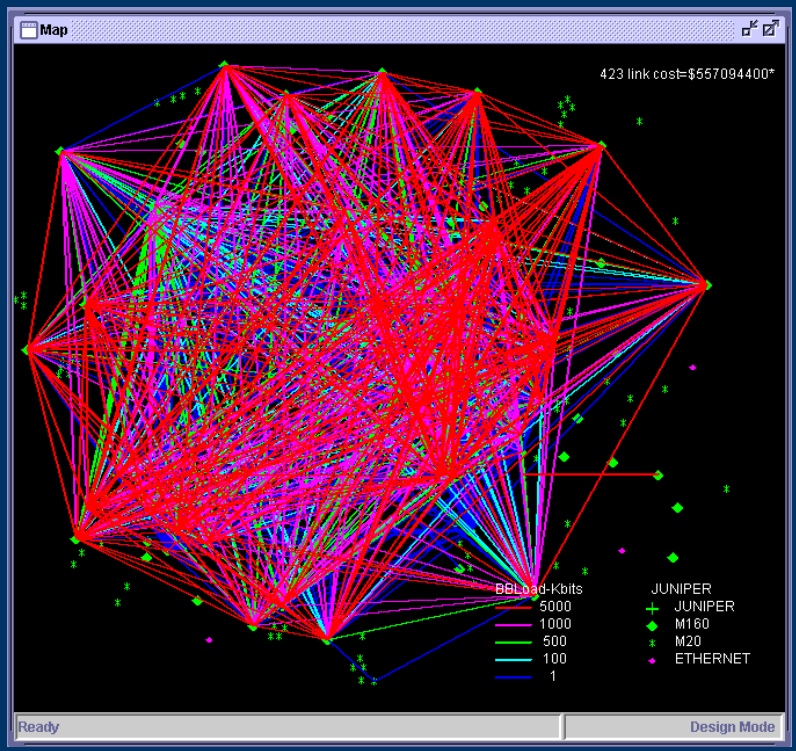


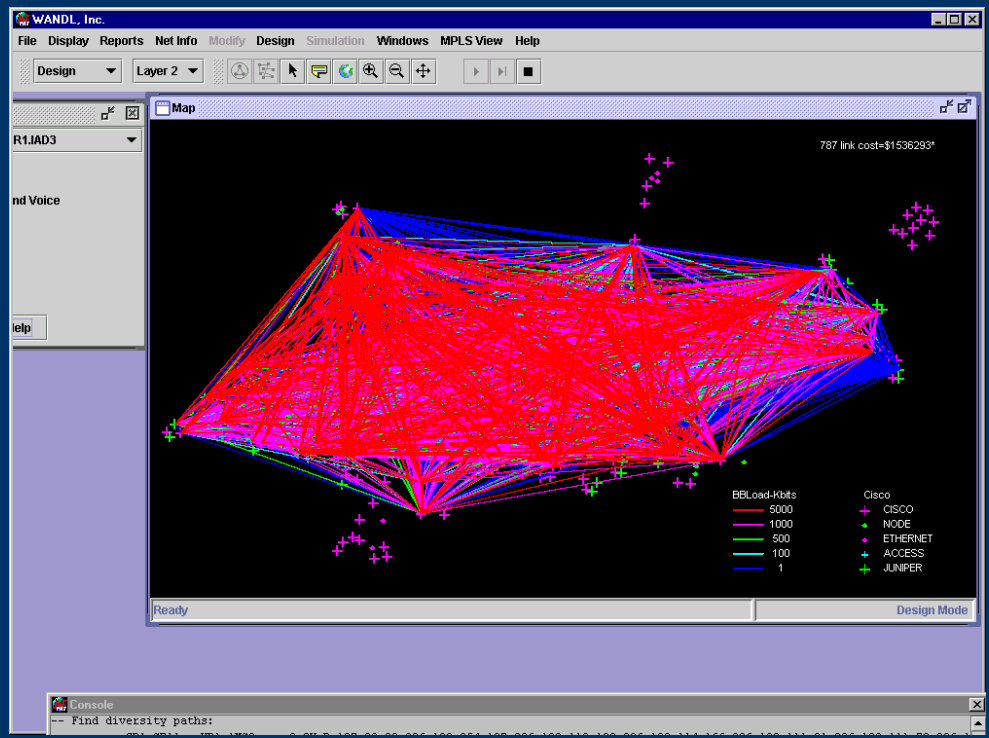
Cisco & Juniper

- Topology Design
- Diversity Design
- Traffic Load Analysis
- Event Simulation
- Failure Analysis
  - Play # Scenario Analysis
- Layer 3 Traffic Performance Analysis & QoS

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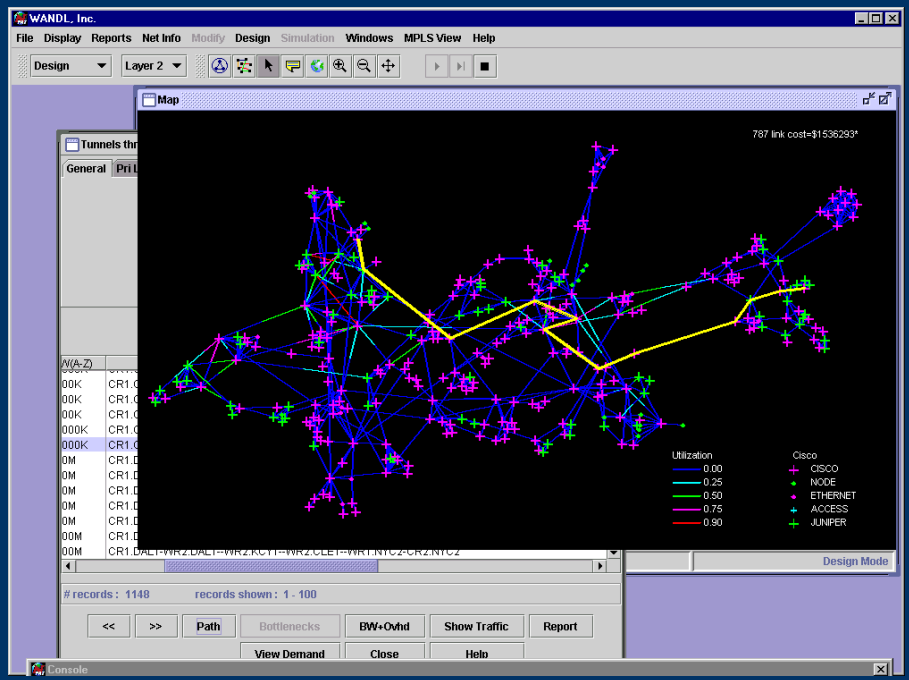
Tunnel Load

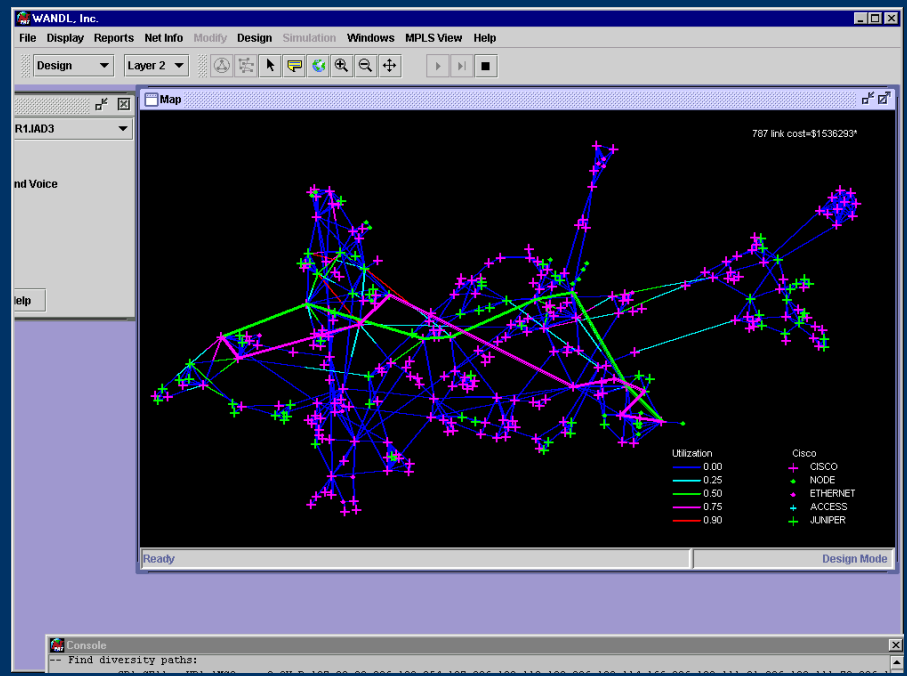




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Primary & Backup LSP Path





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MPLS Traffic Management

- Network Centric
- Network Monitoring and quantifying
  - Characterize traffic and monitor resource utilization
    - Quantifying traffic across the network or its elements in terms of bytes and packets (in and out)
    - Network path, link, LSP
  - Monitor operational status and availability
  - Discover network topology
  - Present physical and logical network topology
  - Monitor network connectivity
  - Pinpoint network integrity problem
  - Provision MPLS tunnels

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Network Status Display

- Semi Real Time Network View
  - View of table/tree structure of networks, areas, hubs/Pops, devices, LSPs and the related attributes
    - Primary attributes: AS number, Area/Level/Region IDs, location, vendors, devices limitation, circuit bandwidth, RSVP Bandwidth,LSP utilization, and LSP path
  - Ability to sync with real time network to update the view
  - Tunnel Path Analysis

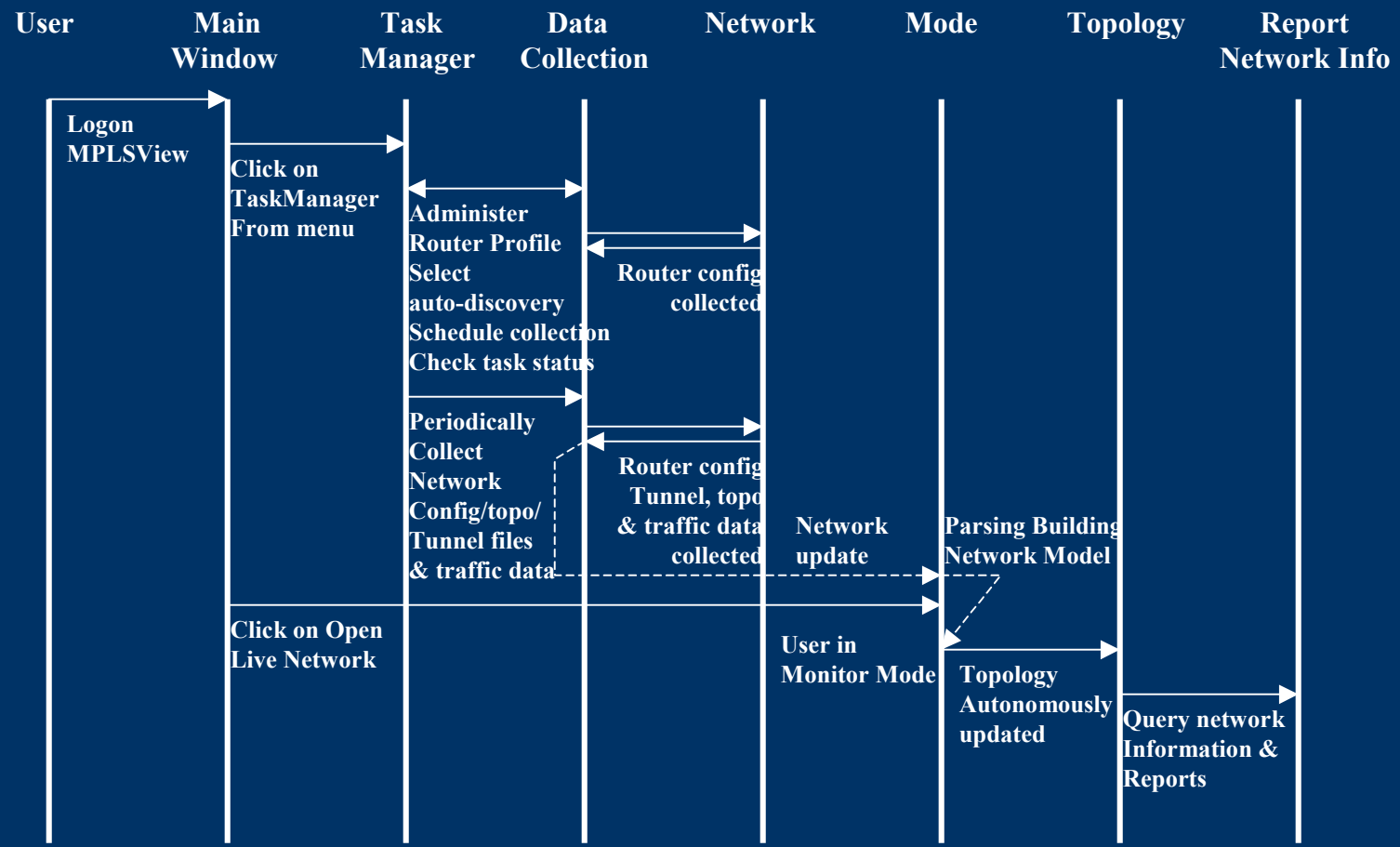
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Network Status Display

- **Network Status Changes**
  - Display/report the updates of network components and their related attributes (new, deleted, and missed trunks, trunk capacity changes), (new LSP, deleted LSP, updated LSP in bandwidth, route changes, priority, info)
- **Network Play Back (historical events)**
  - Replay network status/topology changes
  - Replay the LSPs re-route activities

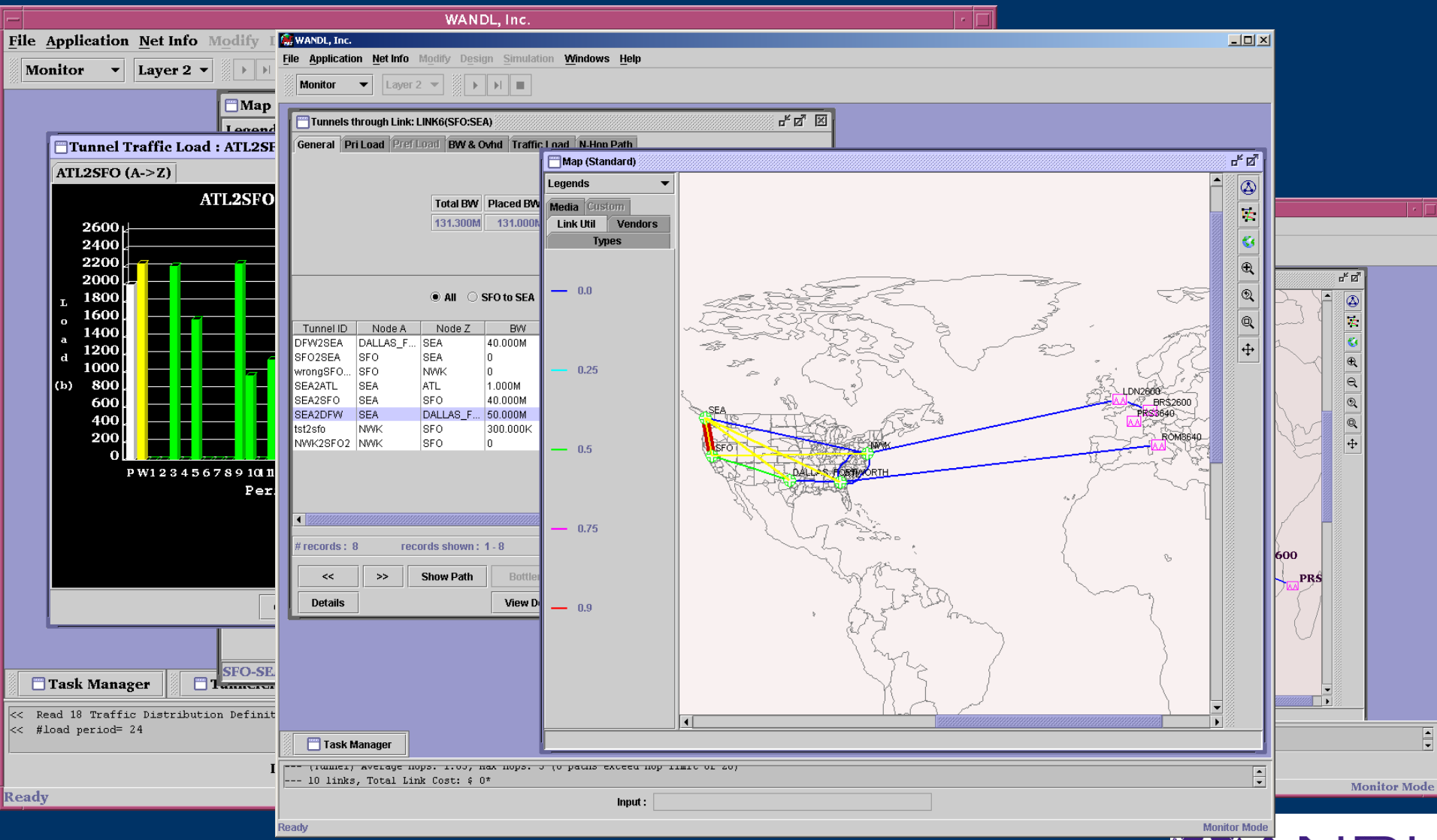
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Monitoring A Live Network



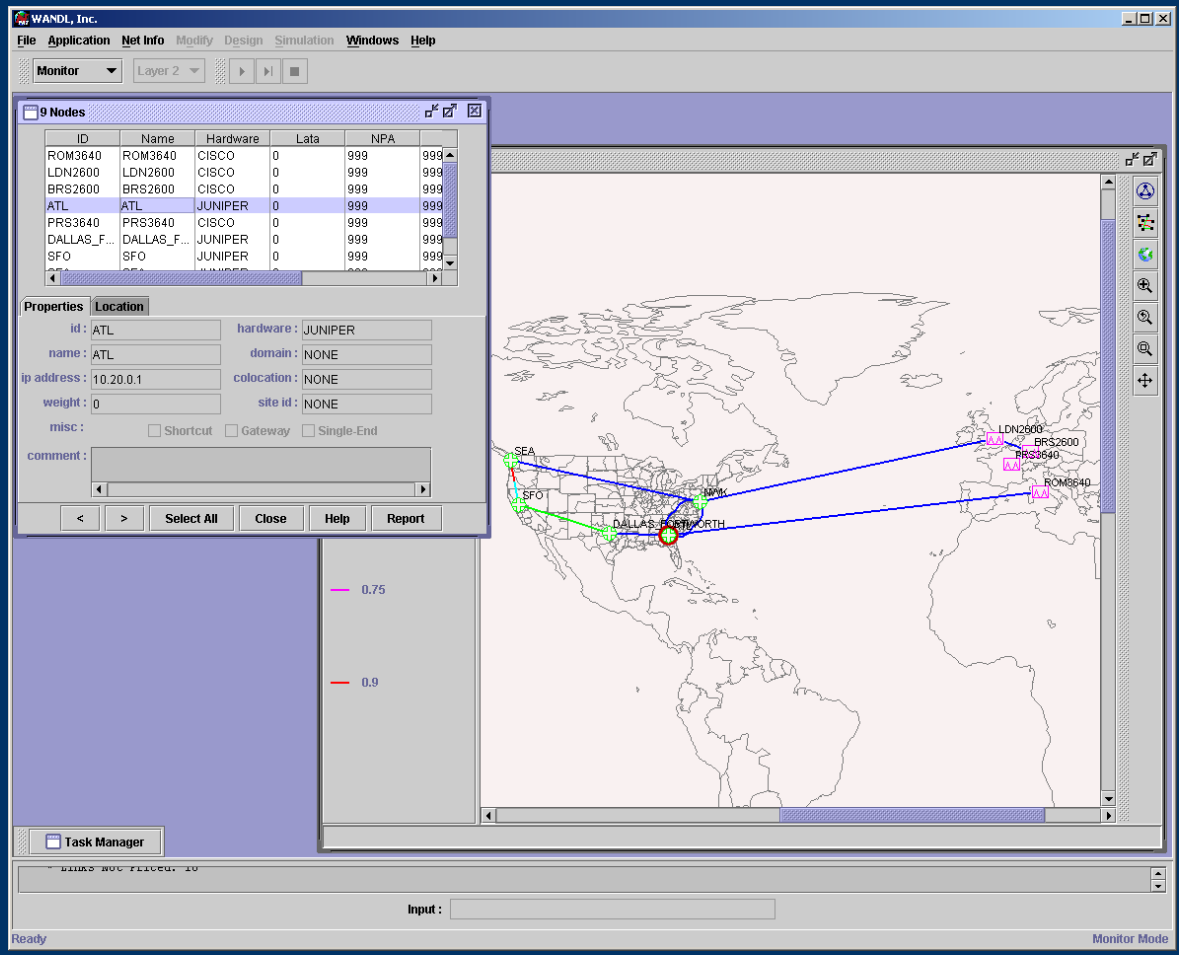
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LSP Traffic Monitoring



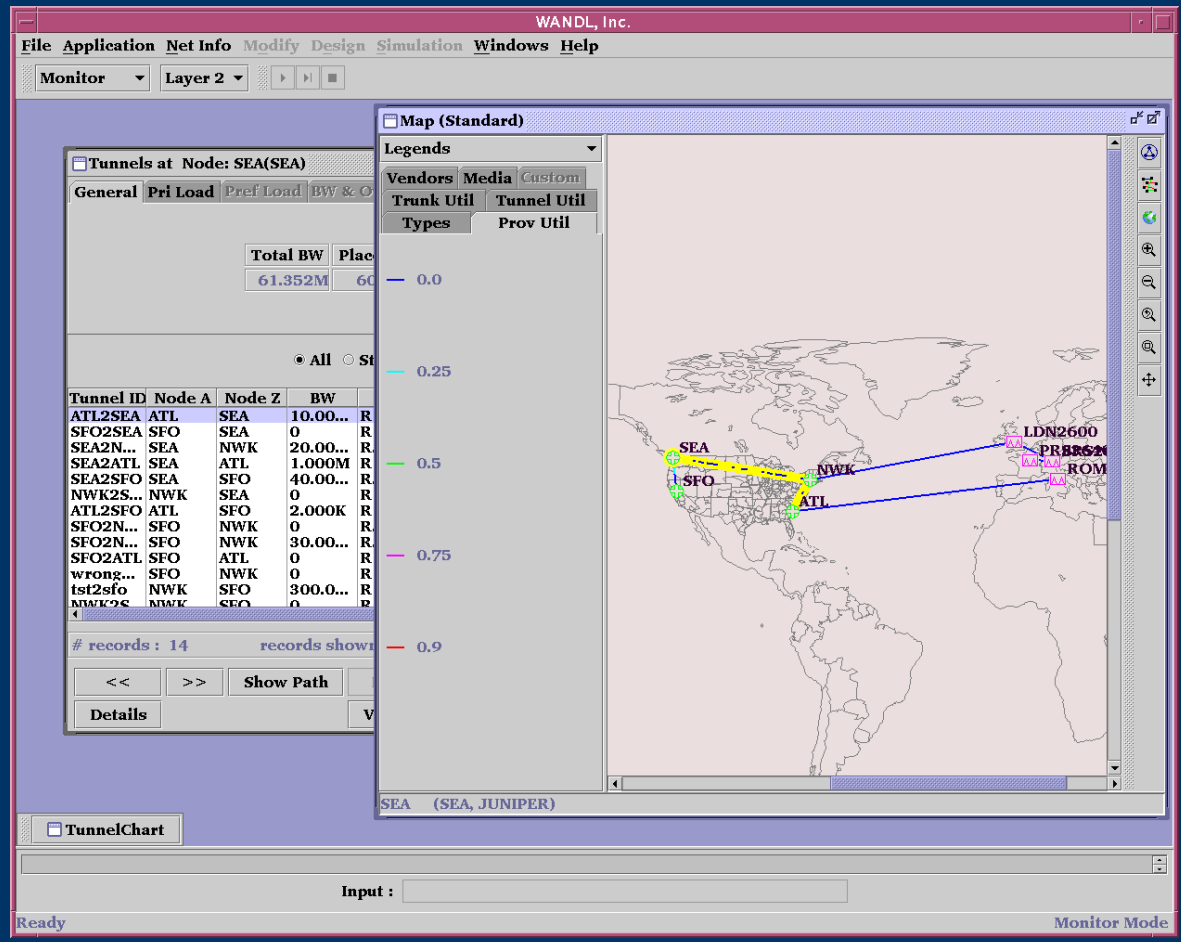
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Network Nodes Information



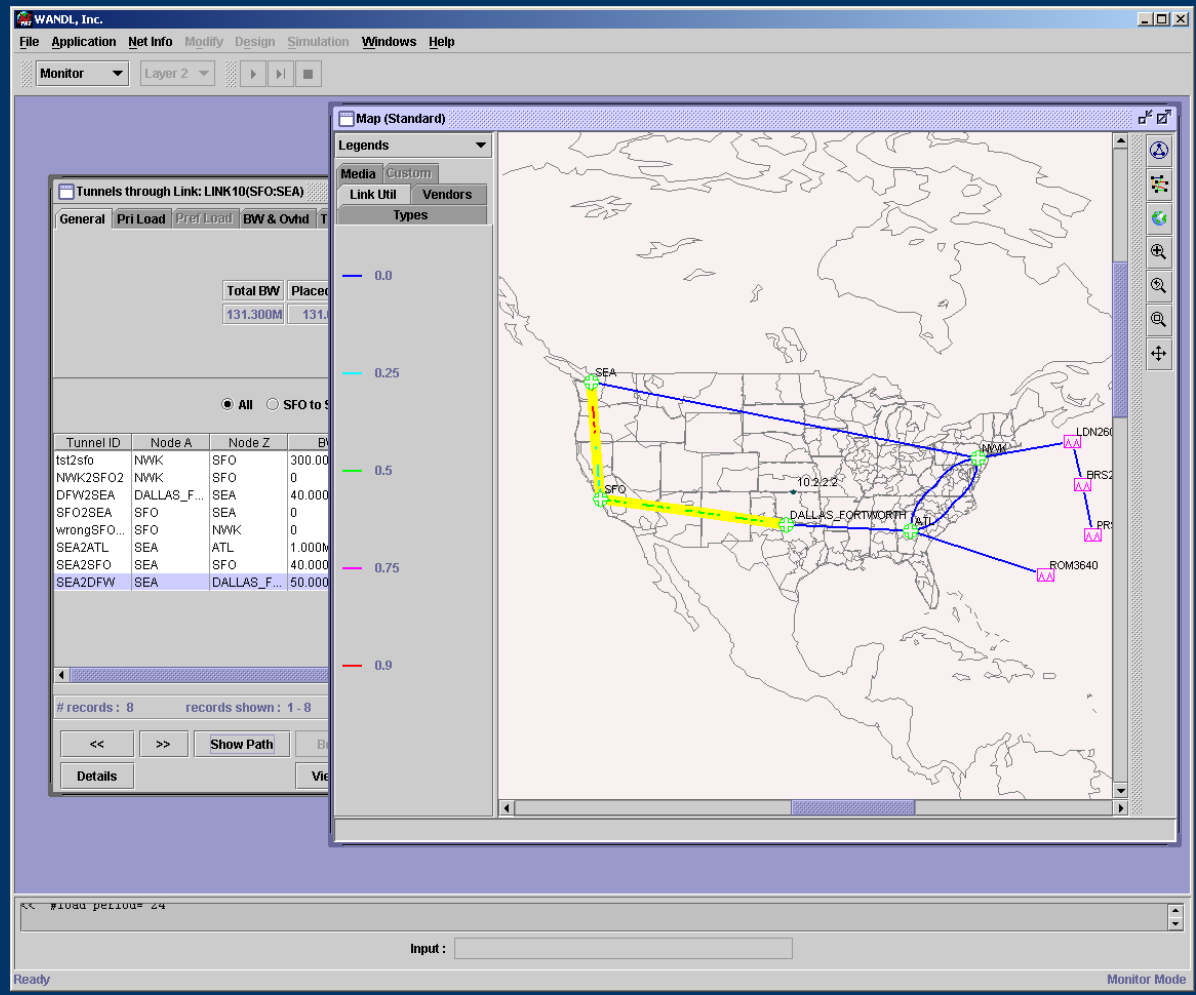
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View Tunnels Thru Node



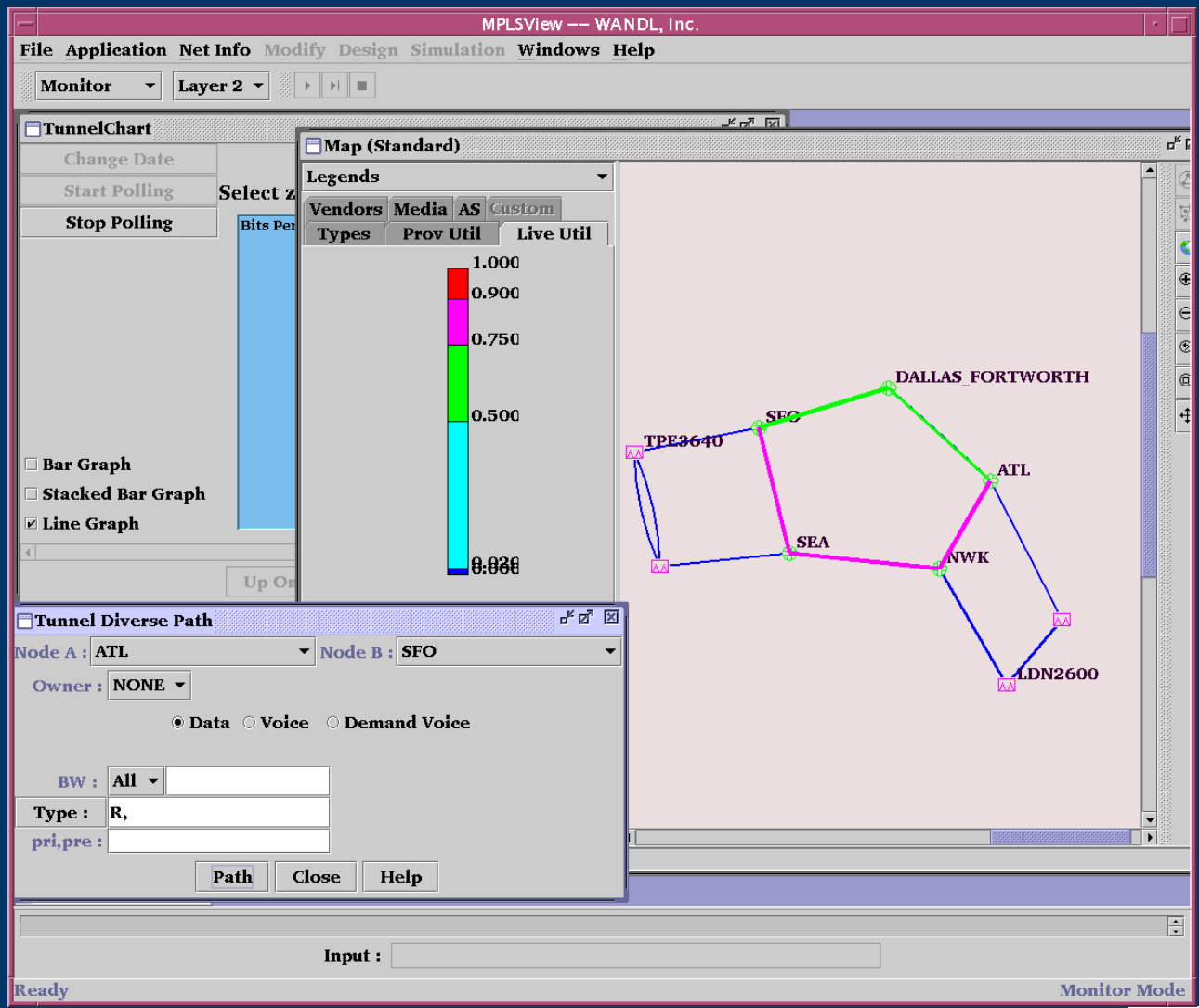
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Display LSP Path Table and on Map



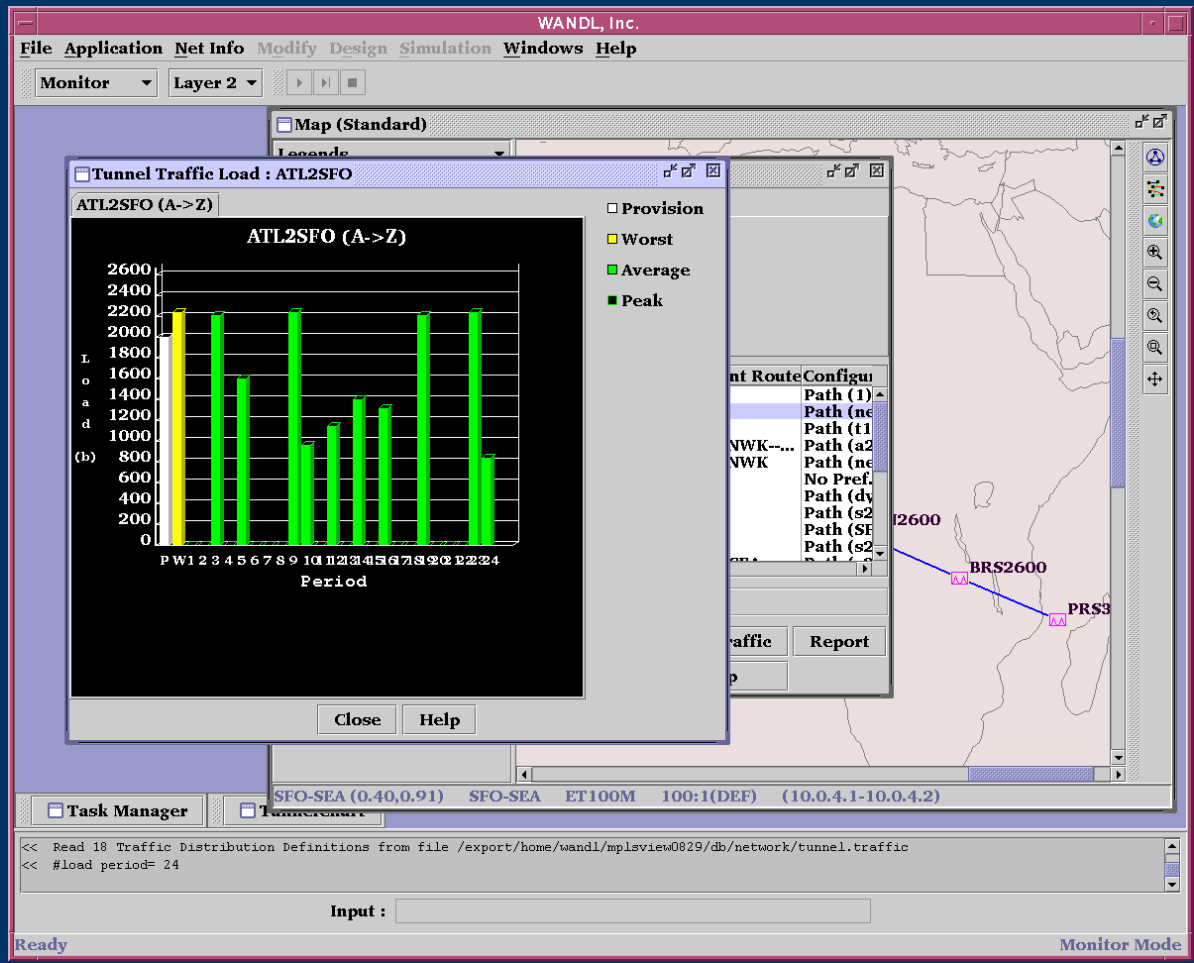
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Primary & Backup Path



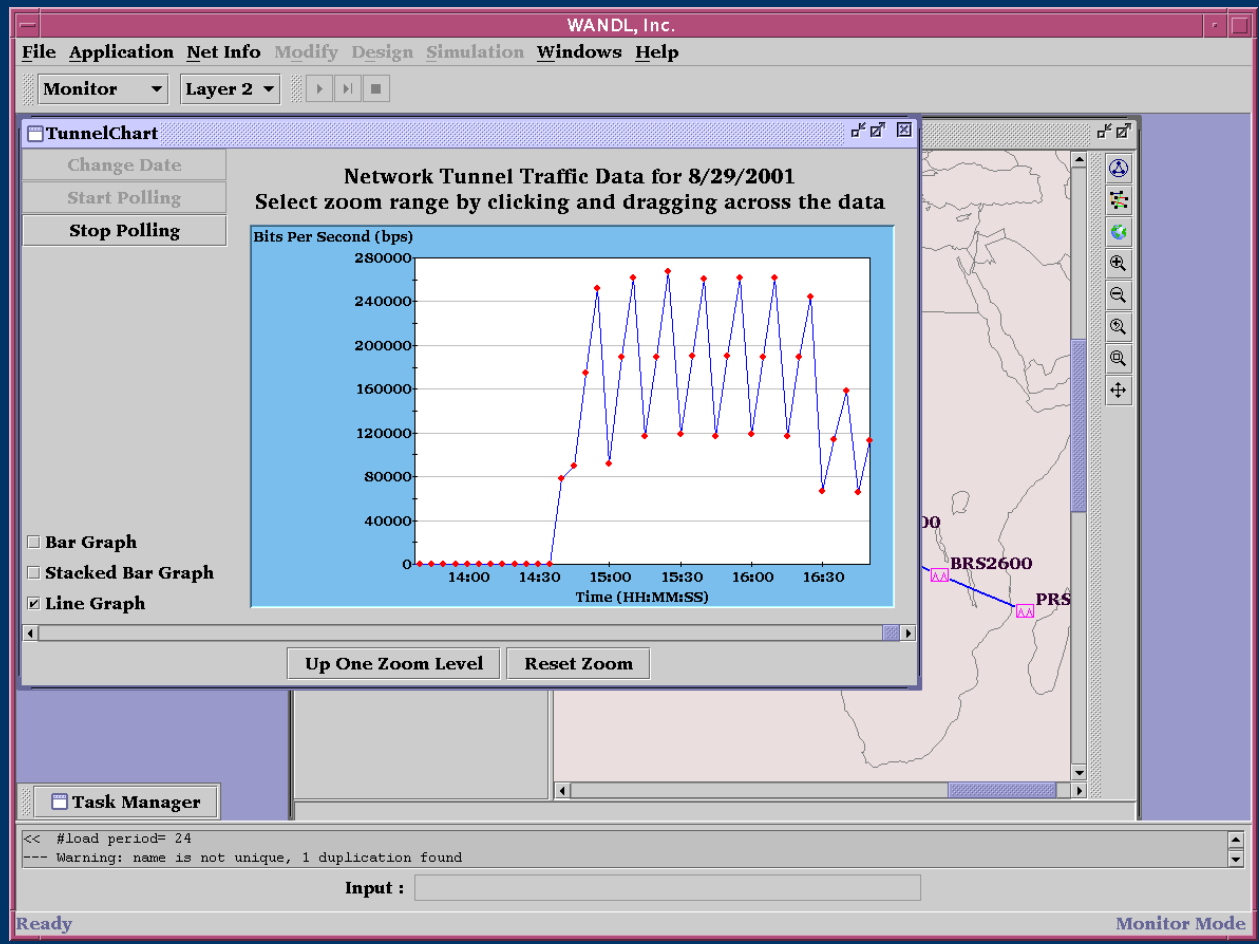
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Tunnel Traffic Load



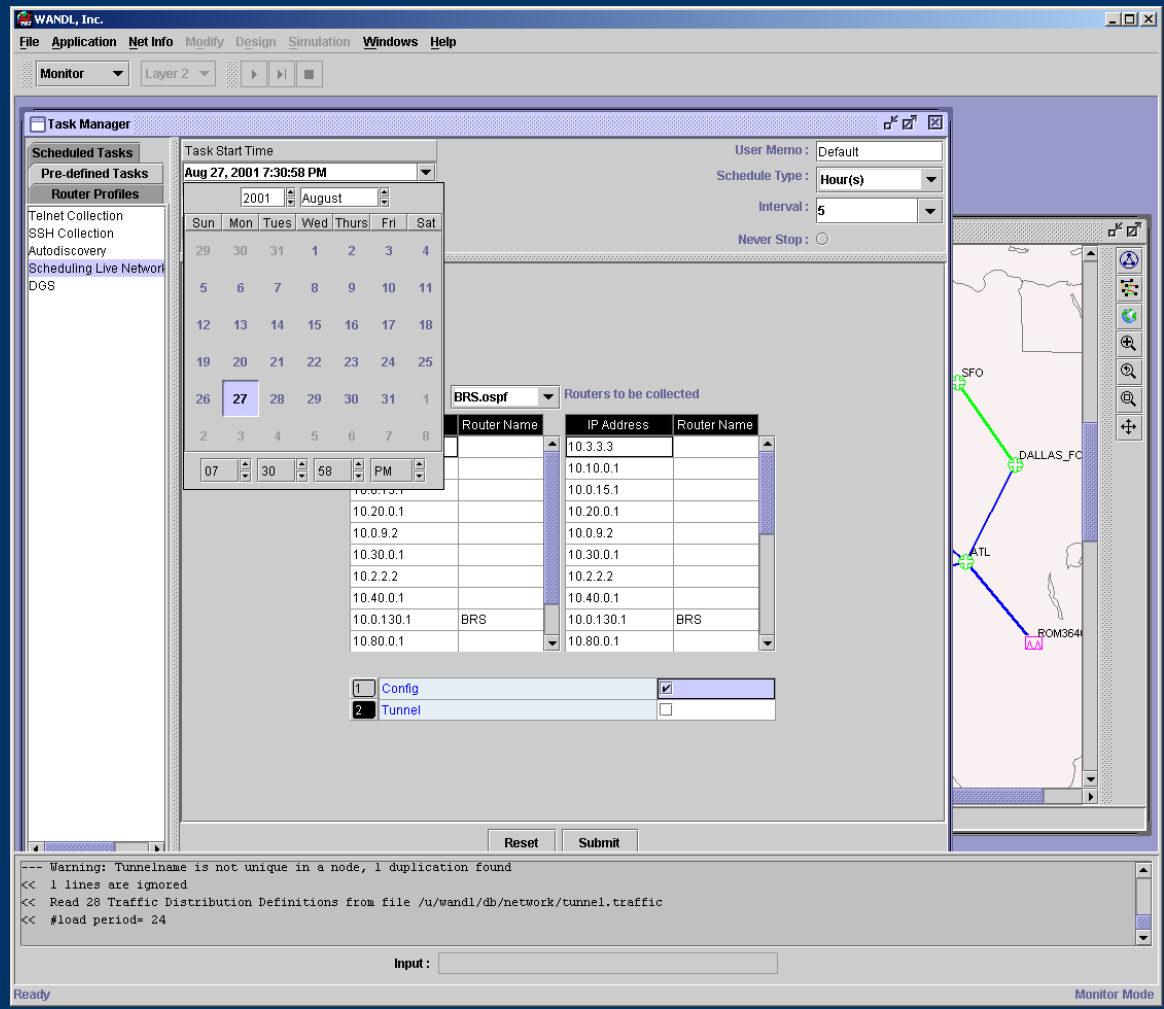
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Aggregated Network Tunnel Traffic



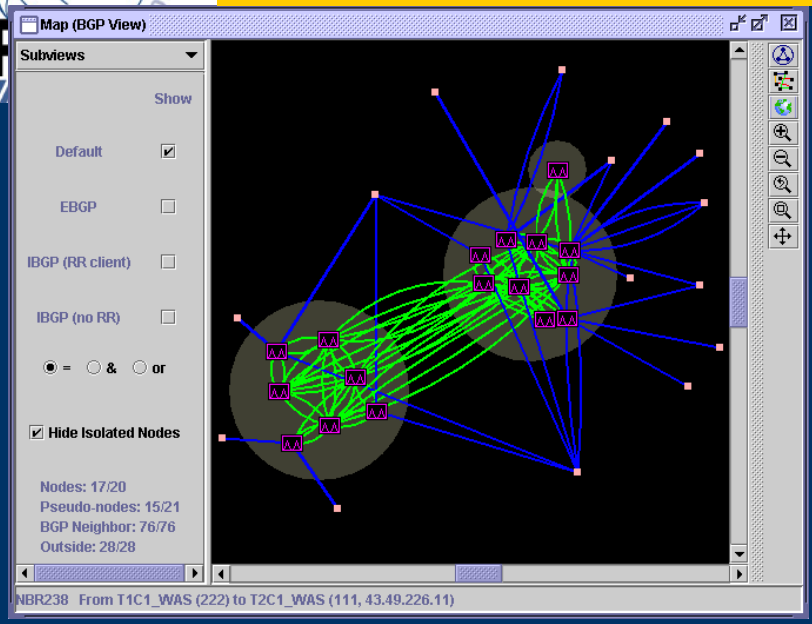
Copyright ©2000, WANDL, Inc.

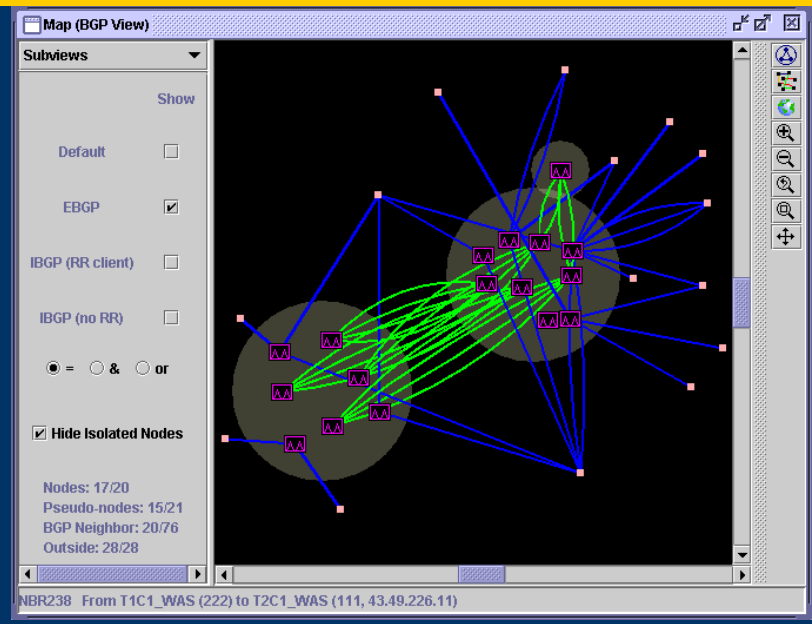
Predefined Tasks GUI – Schedule Data Collection

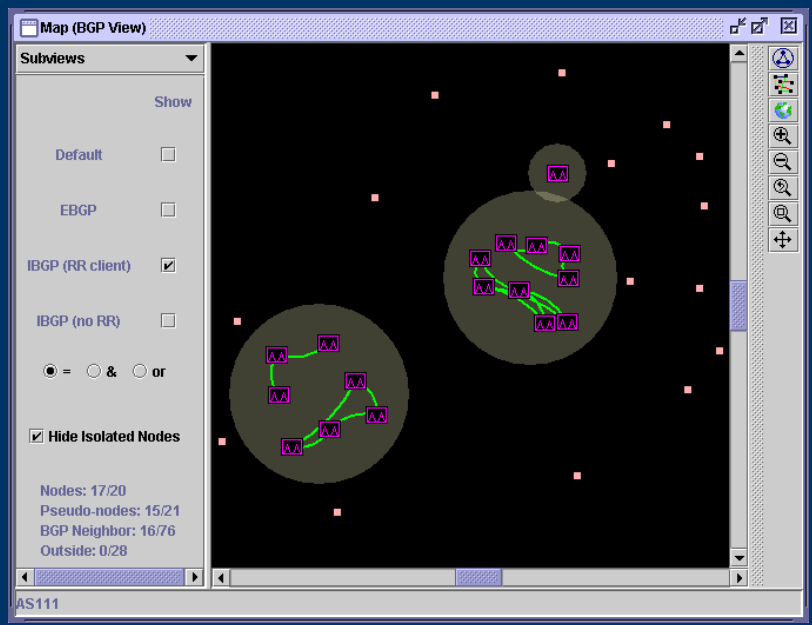


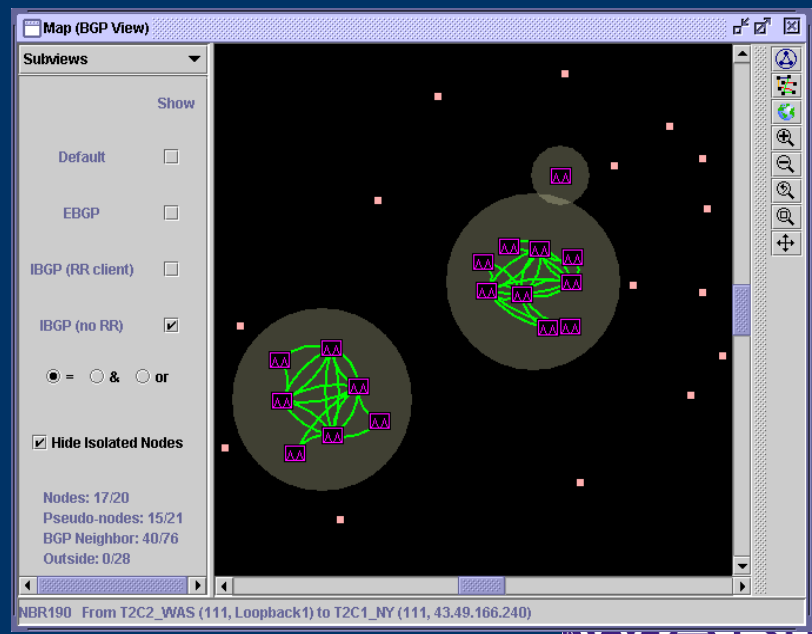
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BGP View



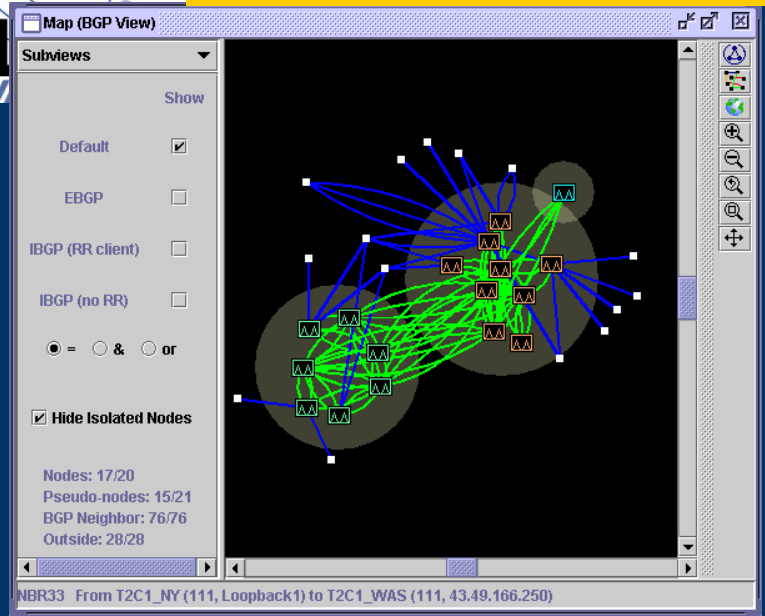


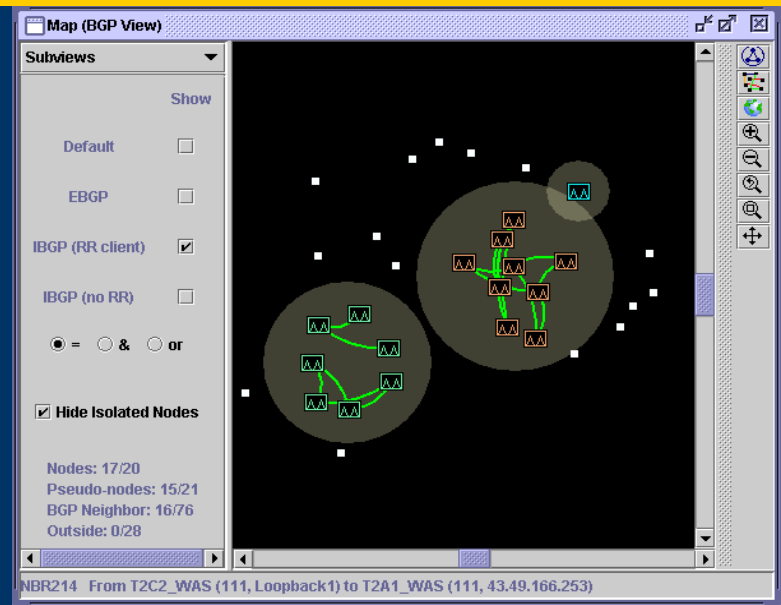


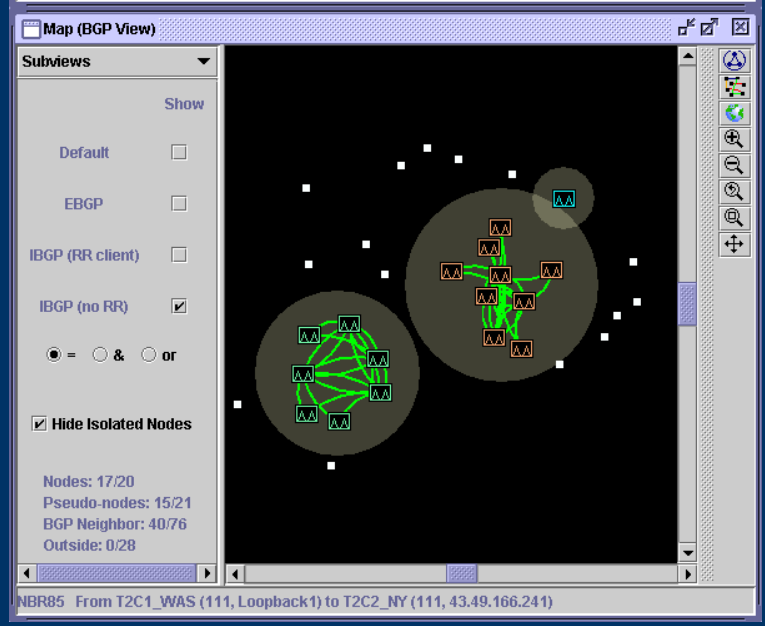


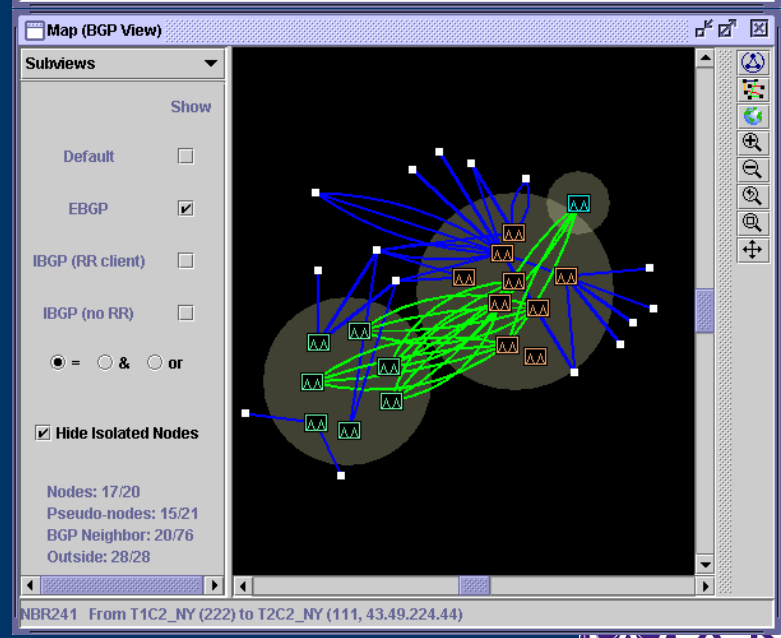
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BGP View



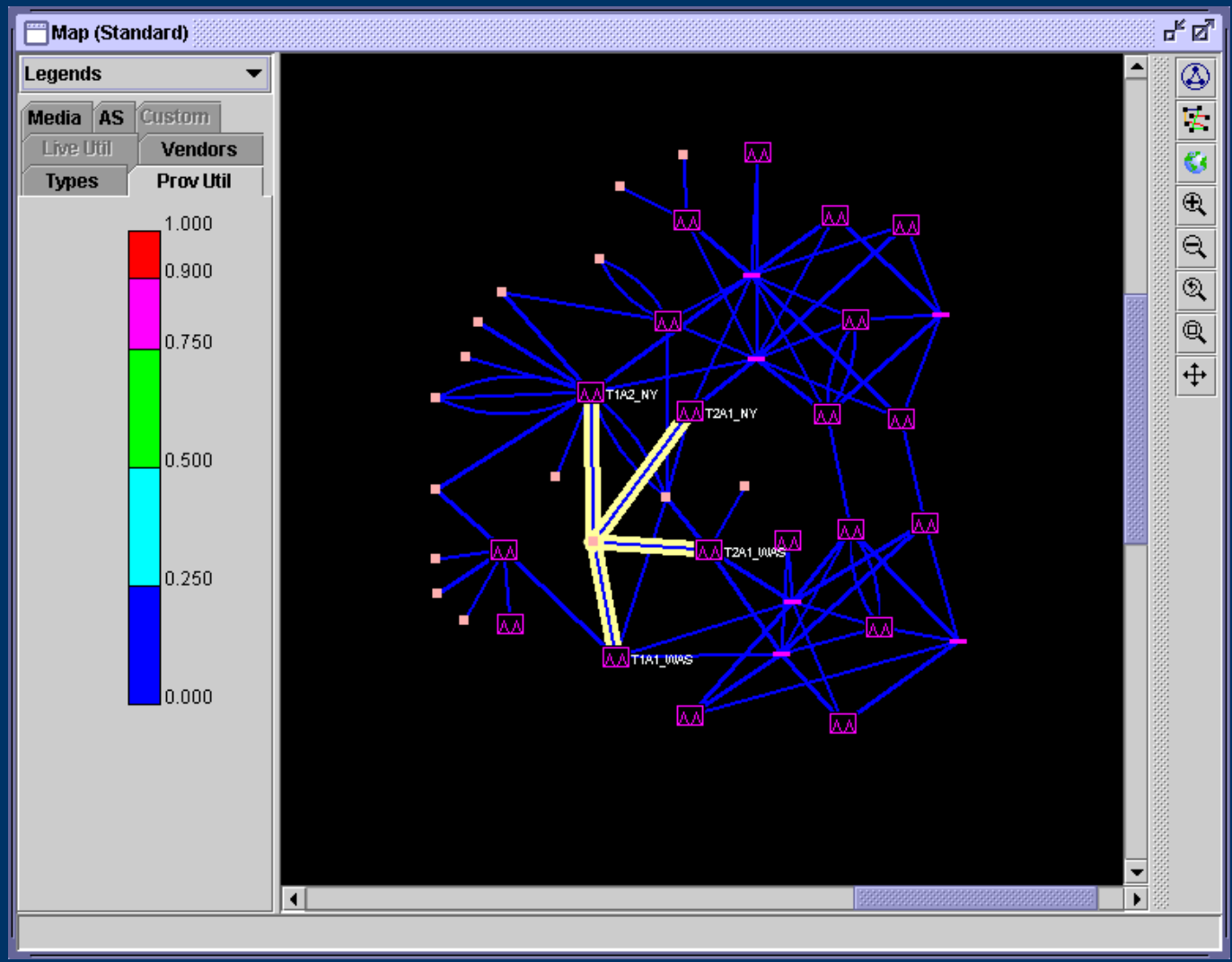






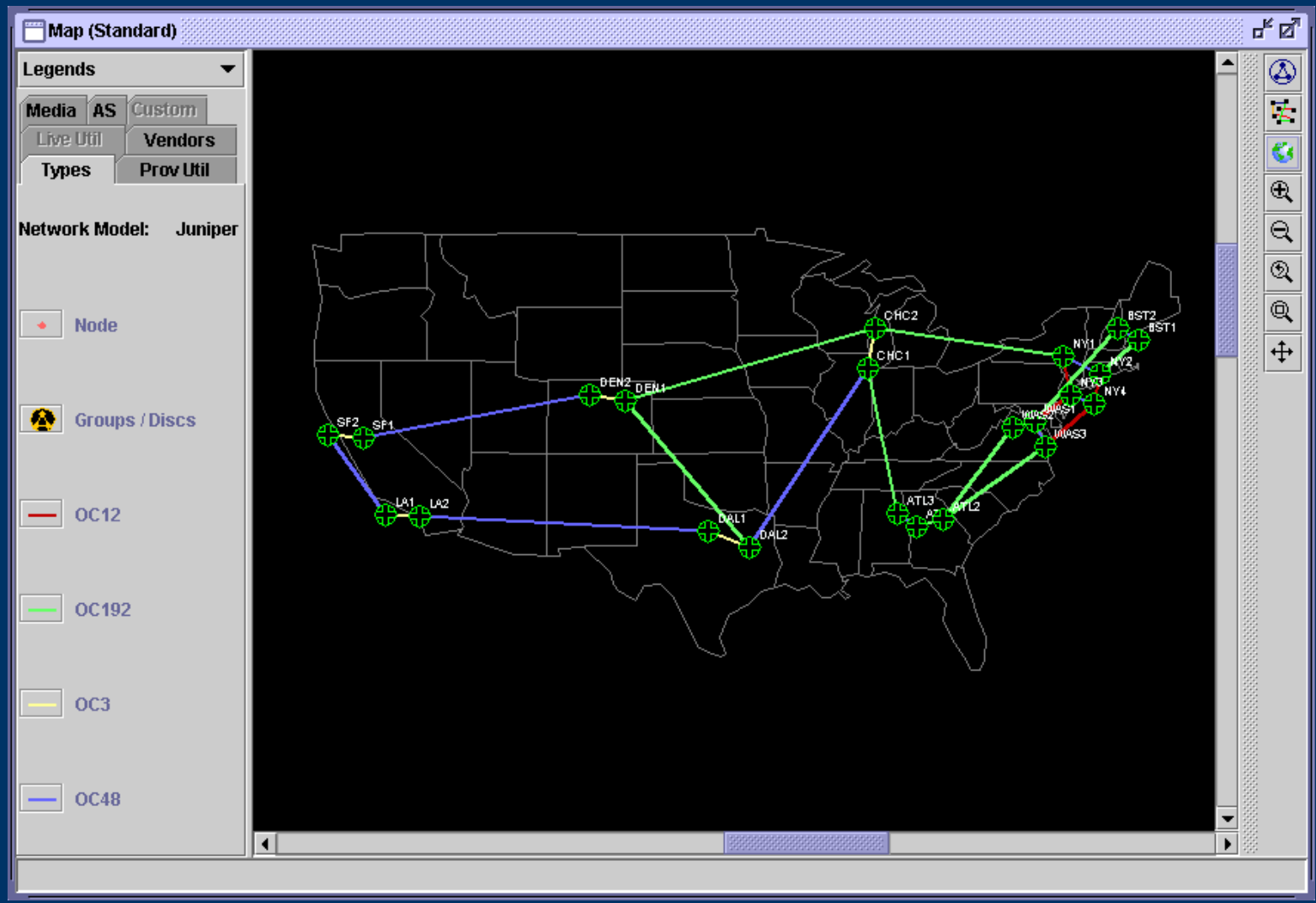
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IP Logical View



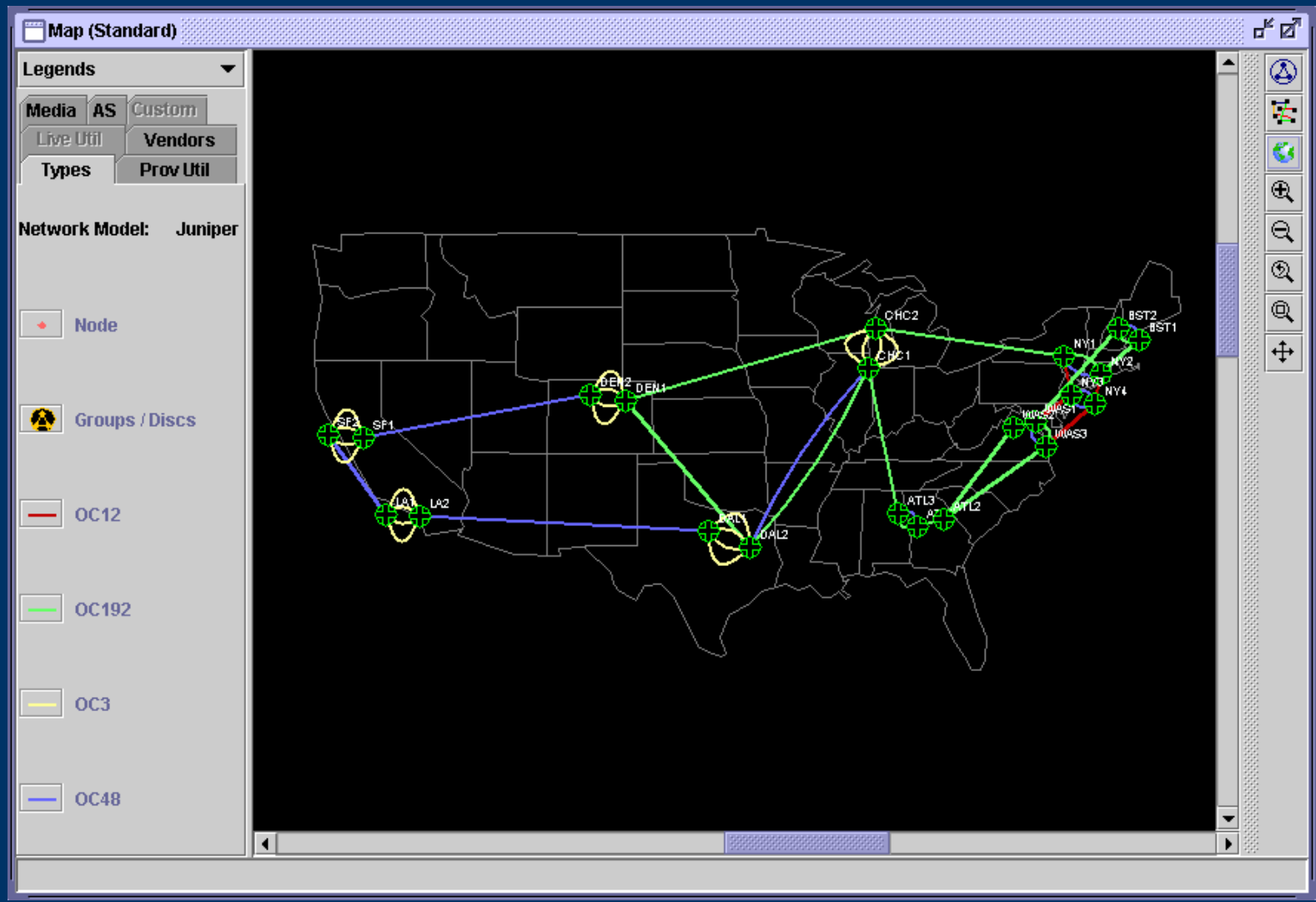
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Transport Layer View



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Multiple Links Between Nodes



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